

PROBLEM STATEMENT

AI-Based Network Intrusion Detection and Threat Monitoring System

Industry Context

In today's digital world, computer networks have become an important part of almost every organization. Educational institutions, hospitals, banks, industries, government organizations, and private companies depend on networks to store information, communicate with users, provide online services, and manage their daily operations. The increasing use of cloud platforms, connected devices, remote access, and online applications has made network communication more important than ever. At the same time, this rapid growth has also increased the number of opportunities for cyber attackers to target organizational networks.

Network attacks can take different forms. Attackers may try to gain unauthorized access to systems, steal confidential information, interrupt services, spread malware, or misuse network resources. Common examples include denial-of-service attacks, port scanning, brute-force attempts, malware-related activities, unauthorized connections, and unusual traffic patterns. Some attacks may happen suddenly, while others may remain hidden for a long period of time. If such activities are not detected at an early stage, they can cause serious damage to the organization.

A network intrusion can result in several problems, such as:

- Data theft and loss of confidential information
- Unauthorized access to systems and network resources
- Service interruptions and network downtime
- Financial losses due to security incidents
- Leakage or misuse of sensitive information
- Damage to the organization's reputation
- Loss of customer trust
- Additional time and cost required for security recovery
- Difficulty in identifying the actual source of an attack

Many organizations use firewalls, antivirus software, intrusion detection tools, and rule-based security systems to protect their networks. These systems are useful for identifying known threats, but traditional approaches may have difficulty detecting new or unusual attack

behavior. They may also generate a large number of alerts, making it difficult for administrators to identify which activities require immediate attention.

Another challenge is the increasing volume of network traffic. Manually checking every connection, packet, and activity is not practical when a network contains a large number of devices and users. Security administrators need a system that can continuously observe network behavior and identify suspicious activities automatically.

Artificial Intelligence and Machine Learning can be used to address this problem by learning patterns from network traffic and identifying activities that differ from normal behavior. An intelligent monitoring system can analyze network data, recognize suspicious patterns, classify possible attacks, and provide alerts to the administrator.

Therefore, there is a need for an AI-based network security solution that provides continuous monitoring, faster threat detection, intelligent analysis, and clear visualization of security events. Such a system can help organizations understand what is happening inside their network and take appropriate action when suspicious activity is detected.

Problem Statement

Design and develop an AI-Based Network Intrusion Detection and Threat Monitoring System that continuously monitors network traffic and system activities to identify potential security threats.

The system should collect and analyze relevant network information such as source and destination addresses, port numbers, communication protocols, packet information, connection duration, traffic volume, and other important network features. The collected data should be processed before analysis so that unnecessary, incomplete, or inconsistent information does not affect the detection process.

The system should use suitable Artificial Intelligence or Machine Learning techniques to analyze network behavior and identify patterns that may indicate an intrusion. It should be capable of detecting abnormal activities as well as different types of known network attacks. When suspicious activity is detected, the system should classify the event and assign an appropriate threat or severity level.

The system should provide real-time alerts to security administrators when important suspicious activities are identified. A monitoring dashboard should display useful information such as the number of detected threats, type of attack, severity level, source of the activity, affected system, and current security status. This will help administrators understand the situation without having to manually examine large amounts of raw network data.

The system should also maintain historical records of detected incidents. These records can be used to understand previous security events, analyze repeated attack patterns, and study changes in network behavior over time. The system should provide simple reports and visualizations so that administrators can easily review network security information.

The proposed solution should be designed to support continuous monitoring and should be capable of handling increasing amounts of network data. It should also consider important requirements such as secure data handling, reliability, scalability, efficient processing, and timely alert generation. The overall aim is to create an intelligent monitoring system that can help detect network intrusions at an early stage and improve the organization's ability to respond to potential security threats.

In addition, the system should provide a clear method for prioritizing detected security incidents based on their severity and potential impact. Low-level suspicious activities should be distinguished from critical threats so that administrators can focus on incidents that require immediate attention. The system should also maintain an audit trail of detected events, including the time of detection, type of activity, source information, severity level, and actions taken by the administrator. This information can be used for reviewing previous incidents, identifying repeated attack patterns, and improving future security monitoring. The system should be designed with proper access control and secure handling of network information to prevent unauthorized access to sensitive security data.

Objectives

The system should:

- To continuously collect network traffic and relevant security data.
- To monitor network activities and connections in real time.
- To process and prepare network data before performing threat analysis.
- To identify unusual and suspicious patterns in network behavior.
- To detect different types of network attacks and intrusion attempts.
- To use AI/ML techniques to improve the identification of potential threats.
- To classify detected activities based on their type and severity.
- To generate real-time alerts when suspicious activities are detected.
- To provide a scalable system that can handle increasing network traffic and security data.
- To improve the organization's ability to identify and respond to potential network threats.

Expected Outcomes

1. Develop an end-to-end AI-Based Network Intrusion Detection and Threat Monitoring System for collecting, processing, analyzing, and monitoring network traffic data.
2. Provide a GUI/dashboard for real-time monitoring, visualization, and analysis of network activities and detected security threats.
3. Maintain information such as source and destination IP addresses, port numbers, protocols, traffic patterns, attack types, threat levels, detection timestamps, and security alerts.
4. Enable AI/ML-based detection and classification of abnormal network behavior and potential intrusion attempts.
5. Provide real-time threat alerts and notifications to help administrators identify and respond to suspicious activities quickly.
6. Maintain secure and efficient storage, retrieval, and management of network traffic data, security events, and historical threat records.
7. Allow users to view, analyze, and export detected threat information and security reports in suitable formats such as CSV, JSON, and report formats.
8. Apply key software engineering principles such as security, scalability, maintainability, reliability, usability, and performance, while demonstrating proper requirements analysis, system design, SDLC selection, risk evaluation, limitations, and future enhancements.